A Natural Language Processing platform for automatic extraction of microbial phenotype information from scientific articles.

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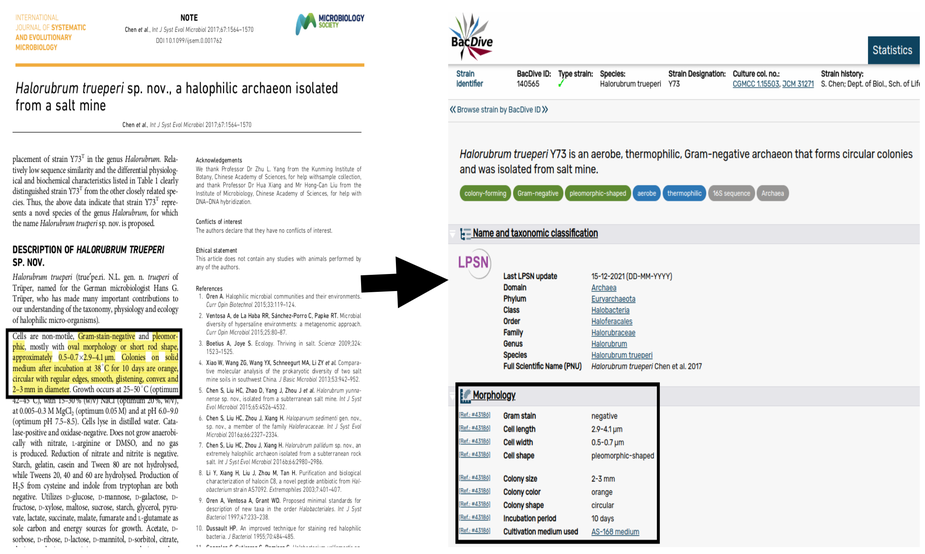
Introduction

Describing bacterial phenotypes is an essential component to identify bacterial species and studying biodiversity and evolution. The large-scale integration and analysis of full phenotypic traits of an organism, can enrich existing data and provide novel scientific insights. The lack of tools for efficiently and accurately extracting information in a structured manner can be attributed to heterogeneity and fragmentation of the data sources, variable standards, and limited interoperability of the data. The project Digital Approaches for the Synthesis of Poorly Accessible Biodiversity Information (DiASPora) aims to establish an automated pipeline for the extraction of phenotypic characters from published literature by application of Natural Language Processing and AI models. In this poster, we describe our progress on the development of the web server for automating data curation and thereby enriching the information available in the database for standardized bacterial information 'BacDive'.

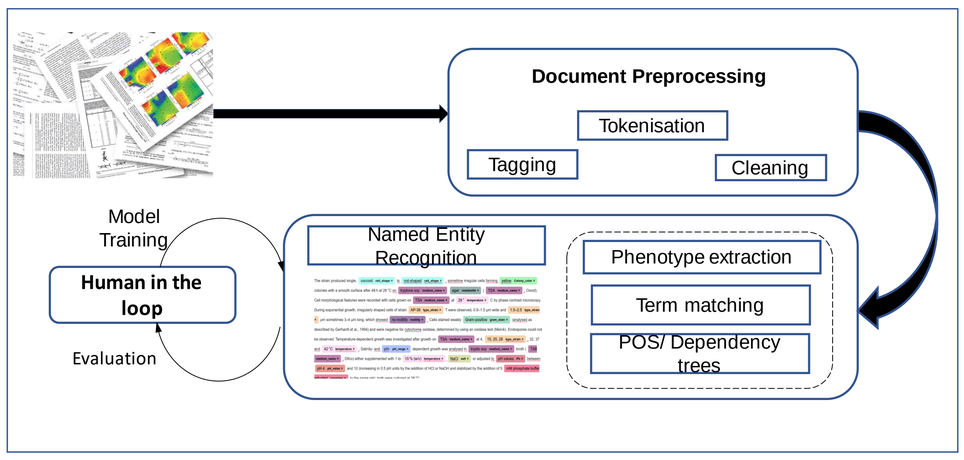
The manual process of information extraction in BacDive



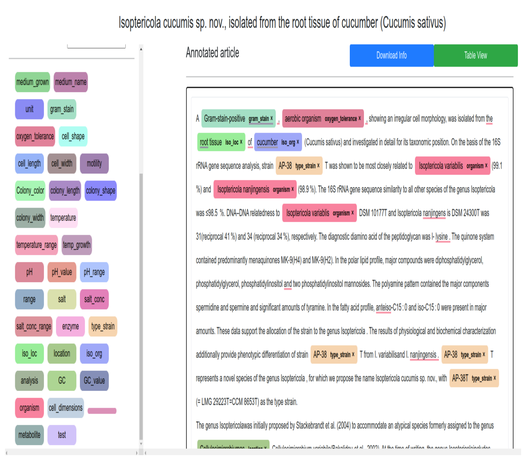
DiASPora NLP system design

- Source of information: The input to the system are freely available full-text articles and stored as xml files for processing.
- Phenotypes to be extracted: Based on feedback from the curators with microbiology background based at DSMZ, 46 phenotypic characters divided broadly into 8 categories are being extracted.
- Information extraction: Information extraction consists of the following steps:
  - ScispaCy model trained on the Joint Workshop on Natural Language Processing Biomedicine and its Application (JNLPBA) corpus is used for sentence splitting.
  - For each phenotype, a spaCy model was trained using sentences which were annotated with phenotypes and their respective values as provided by experts.
  - The trained model is used to annotate sentences of an article in runtime.
  - The annotated sentences will be passed into appropriate phenotype extractors, which are utilize the tags for tabulating the information.

The system for information extraction



Example extraction from the article by Kämpfer, et. al., 2016, Int J Syst Evol Microbiology, https://doi.org/10.1099/ijsem.0.001055



Phenotypic Character	Value
Gram stain	Gram-stain positive
Oxygen tolerance	Aerobic
Isolation location	Root tissue
Isolation organism	Cucumber
Type strain	AP38T
Organism	Isoptericola nanjingensis, Isoptericola variablis

Salient features

- Articles from 12 journals supported
- Runtime annotation of articles
- Over 6000 articles indexed
- Annotation tool with multiple features:
  - The ability to add and remove annotations and save the new annotations.
  - Annotated text can be downloaded as a JSON file.
  - The approved / manual annotations used for training the model

Future Work

- Improvement in the phenotype extractor to represent the BacDive data format.
- Integrate the ontology being developed by our partners at TIB with the annotations.
- Store information in form of knowledge graphs for better usability and knowledge discovery.

Scan to find out more

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Partners

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Leibniz Institute DSMZ-German Collection of Microorganisms and Cell Cultures GmbH

TIB Leibniz-Informationszentrum Technik und Naturwissenschaften Universitätsbibliothek

Leibniz Gemeinschaft